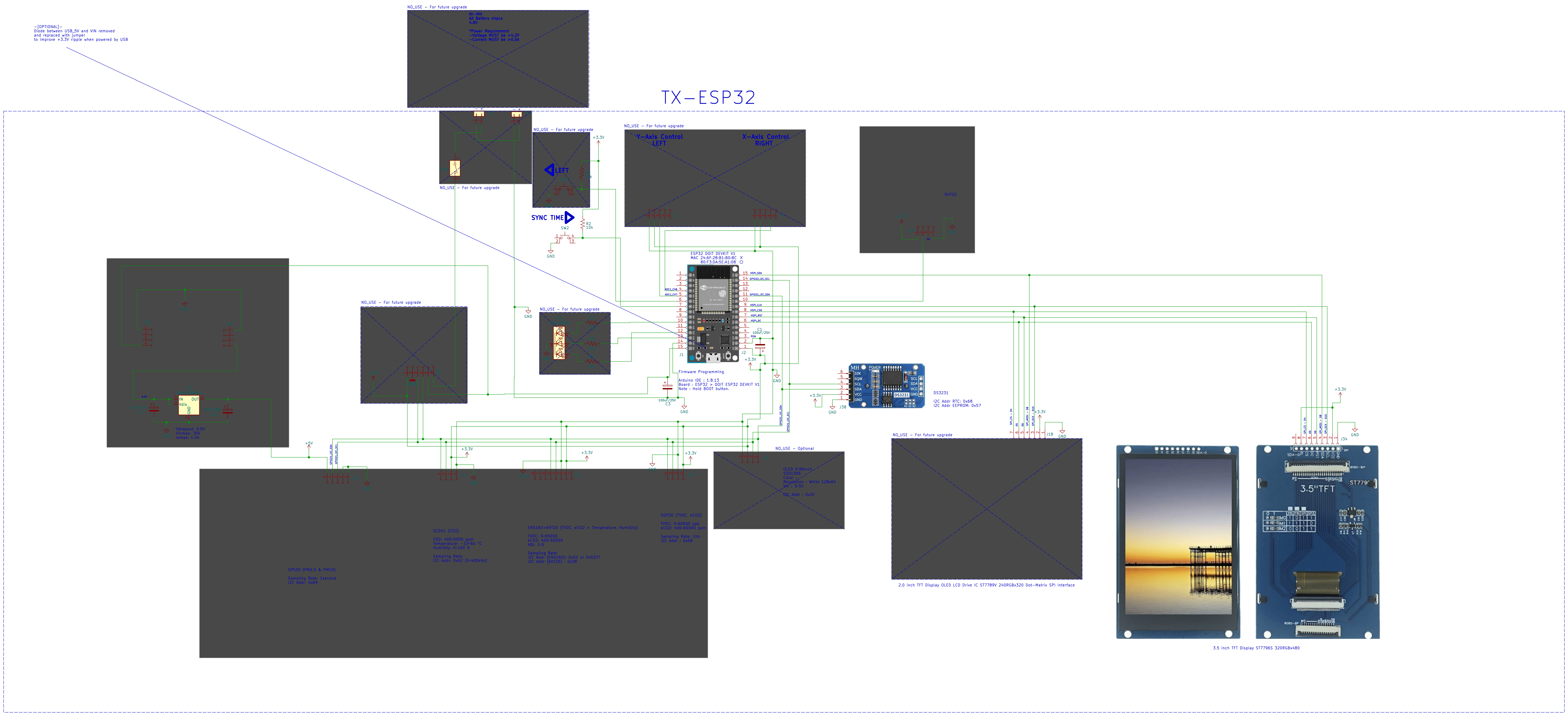
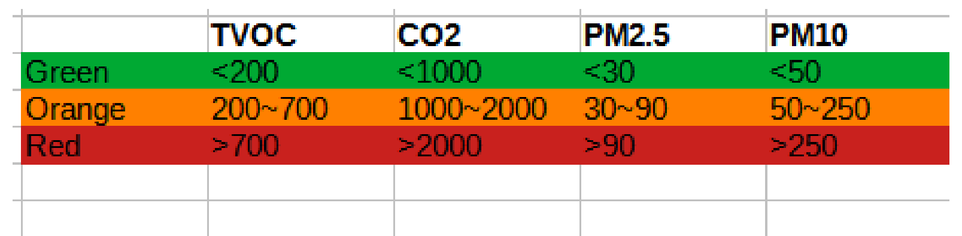
-[OPTIONAL]-
Diode between USB_5V and VIN removed
and replaced with jumper
to improve +3.3V ripple when powered by USB

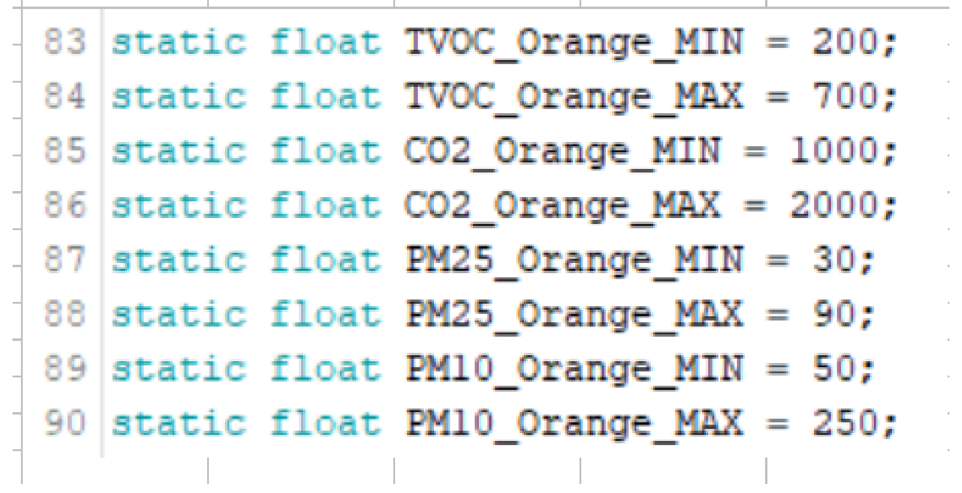

NO_USE - For future upgrade
Ni-MH
AA Battery x4pcs
4.8V
*Power Requirement
-Voltage MUST be >4.2V
-Current MUST be >0.5A
TX-ESP32
NO_USE - For future upgrade
LEFT
SW1
R1
10k
+3.3V
SYNC TIME
SW2
R2
10k
GND
NO_USE - For future upgrade
Y-Axis Control
LEFT
X-Axis Control
RIGHT
J3
J4
SHT31
ESP32 DOIT DEVKIT V1
MAC 24:6F:28:B1:B0:8C X
80:F3:DA:5E:A1:08 O
ADC1_CH6
ADC1_CH7
HSPI_SDA
GPIO22_I2C_SCL
GPIO21_I2C_SDA
HSPI_CLK
HSPI_CS0
HSPI_RST
HSPI_DC
C1
100uF/25V
PON
J1
J2
Firmware Programming
Arduino IDE : 1.8.13
Board : ESP32 > DOIT ESP32 DEVKIT V1
Note : Hold BOOT button.
100uF/25V
C3
DS3231
I2C Addr RTC: 0x68
I2C Addr EEPROM: 0x57
J38
32K
SQW
SCL
SDA
VCC
NO_USE - For future upgrade
NO_USE - For future upgrade
U1
6.0V
IN
OUT
Vdis
100uF/25V
C5
C6
Vdropout: 0.5V
Vinmax: 35V
Iomax: 1.5A
+5V
J13
J12
J11
J10
J9
SPS30 (PM2.5 & PM10)
Sampling Rate: 1second
I2C Addr: 0x69
SCD41 (CO2)
CO2: 400~5000 ppm
Temperature: -10~60 °C
Humidity: 0~100 %
Sampling Rate:
I2C Addr: 0x62 (0~400kHz)
ENS160+AHT20 (TVOC, eCO2 + Temperature, Humidity)
TVOC: 0~65000
eCO2: 400~65000
AQI: 1~5
Sampling Rate:
I2C Addr (ENS160): 0x52 or 0x53??
I2C Addr (AHT20) : 0x38
SGP30 (TVOC, eCO2)
TVOC: 0~60000 ppb
eCO2: 400~60000 ppm
Sampling Rate: 1Hz
I2C Addr : 0x58
NO_USE - Optional
OLED 0.96Inch
SSD1306
Color :
Resolution : White 128x64
Vin : 3~5V
I2C Addr : 0x3C
NO_USE - For future upgrade
SPI_CS - D4
D5
D6
SPI_MOSI - D8
SPI_SCK - D10
J18
J34
3.5"TFT
ST7796S

2.0 inch TFT Display OLED LCD Drive IC ST7789V 240RGBx320 Dot-Matrix SPI Interface

3.5 inch TFT Display ST7796S 320RGBx480



	TVOC	CO2	PM2.5	PM10
Green	<200	<1000	<30	<50
Orange	200~700	1000~2000	30~90	50~250
Red	>700	>2000	>90	>250



```
static float TVOC_Orange_MIN = 200;
static float TVOC_Orange_MAX = 700;
static float CO2_Orange_MIN = 1000;
static float CO2_Orange_MAX = 2000;
static float PM25_Orange_MIN = 30;
static float PM25_Orange_MAX = 90;
static float PM10_Orange_MIN = 50;
static float PM10_Orange_MAX = 250;
```

-[SENSORS MEASUREMENT RANGE]-

SPS30	PM(MC)	0~1000	µg/m3
SPS30	PM(NC)	0~3000	#/cm3
SCD41	CO2	400~5000	ppm
ENS160	TVOC	0~65000	ppb
ENS160	eCO2	400~65000	ppm
ENS160	AQI	1~5	
SGP30	TVOC	0~60000	ppb
SGP30	eCO2	400~60000	ppm

CO2

0 ~ 400 : Normal Outdoor - Green
400 ~ 1000 : Typical Indoor - Green
1000 ~ 2000 : Poor Air - Orange
2000 ~ 5000 : - Red

RGBA Color
Blue (0,0,255,.3)
Red (255,0,0,.3)

Sheet: /
File: KiCad_Air_Quality_Sensor.kicad_sch
Title:
Size: A1
Date:
Rev:
KiCad E.D.A. eeschema 7.0.1
Id: 1/1